CLINICAL TRIAL FAILURES · MISSED ENDPOINTS · RESPIRATORY PHASE 2,3 · STARTS 2015–2024

# IL-13: 4 of 4 completed trials missed their primary endpoint

**4 completed trials across 3 sponsors and 3 molecules ran to the end and missed their primary endpoint on the sponsor's own posted result; 0 met it.** Only 0 of 5 closed trials were stopped early for efficacy or safety, which is why this class is read through its completed trials rather than a stop rate. Most completed trials post nothing readable, so the count is a floor.

**4 of 4**
completed trials with a readable primary result missed it · 0 met it

**3 / 3**
sponsors / molecules behind the misses

**0.0%**
stopped early: 0 of 5 closed trials (95% CI 0.0%–43.4%)

**83.3%**
of 6 trials have closed · 1 still open or unresolved

**4 of 4** **Completed and missed the primary endpoint.** Of 4 completed trials in this segment, 4 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 4 missed and 0 met. Counted separately and never in the rate above; a floor, not a rate. 0 of the 4 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

## The completed trials that missed

| Trial | Ph | Sponsor | Experimental drugs | Started | What the posted result says |
|---|---|---|---|---|---|
| NCT02449473 | 2 | AstraZeneca | TRALOKINUMAB | 2015-09 | Change From Baseline to Week 12, Expressed as a Ratio, in Number of Airway Submucosal Eosinophils — p 0.3862, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Least square (LS) geometric mean ratio 1.43 (95% CI 0.63 to 3.27). Method: ANCOVA. |
| NCT02546700 | 2 | Roche | LEBRIKIZUMAB | 2015-09 | Change From Baseline in Pre-bronchodilator Forced Expiratory Volume in One Second (FEV1) at Week 12 — p 0.867, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Adjusted Mean Difference 5.3 (95% CI -57 to 67.6). Method: Mixed Models Analysis. |
| NCT02345070 | 2 | Sanofi | ROMILKIMAB | 2015-05 | Absolute Change From Baseline in Percent Predicted Forced Vital Capacity (FVC) at Week 52 — p 0.6339 against the sponsor's stated threshold of 0.05: not significant. Least Square (LS) Mean Difference -0.5 (95% CI -2.56 to 1.56). Method: Mixed Models Analysis. |
| NCT02281357 | 3 | AstraZeneca | TRALOKINUMAB | 2015-02 | Percent Change From Baseline in the Final Daily, Average, OCS Dose at Week 40 While Not Losing Asthma Control. — p 0.271, no threshold stated; 0.1 or above, so not significant under any conventional threshold. LS Mean difference -7.78 (95% CI -21.7 to 6.15). Method: ANCOVA. |

## How this compares

IL-13
0.0%
0/5 closed

all respiratory trials with a resolved drug target
4.2%
35/843 closed

All respiratory Phase 2,3
4.2%
82/1946 closed

**What is in this class:** 3 distinct experimental drugs across 6 trials (antibody 6). Most tested: TRALOKINUMAB (3), LEBRIKIZUMAB (2), ROMILKIMAB (1). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 0 stops came from 0 sponsor-asset programmes across 0 sponsors. Screened alongside 13 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. One further closed trial here was terminated with no cause recorded in the registry; if it were biological the rate would be 20.0%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI |
|---|---|---|---|---|
| 2015–2017 | 0 | 5 | 0.0% | 0.0%–43.4% |

## Where the stops sit

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Sponsors with most stops: .

**Method**
Denominator: ClinicalTrials.gov interventional Phase 2,3 respiratory trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 41% of industry respiratory trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 83.3% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**The whole cohort, trial by trial.** The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta: enter your work email at **clinicaltrialfailures.com/asset-check** and it opens straight away.

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.